- Charts: visualizing data
- Getting started with BigQuery

Machine Learning

These are a few of the notebooks related to Machine Learning, including Google's online Machine Learning course. See the full course website for more.

- Intro to Pandas DataFrame
- Intro to RAPIDS cuDF to accelerate pandas
- Getting Started with cuML's accelerator mode

Using Accelerated Hardware

- Train a CNN to classify handwritten digits on the MNIST dataset using Flax NNX API
- Train a Vision Transformer (ViT) for image classification with JAX
- Text classification with a transformer language model using JAX

Featured examples

- Train a miniGPT language model with JAX AI Stack
- LoRA/QLoRA finetuning for LLM using Tunix
- Parameter-efficient fine-tuning of Gemma with LoRA and QLoRA
- Loading Hugging Face Transformers Checkpoints
- 8-bit Integer Quantization in Keras
- Float8 training and inference with a simple Transformer model
- Pretraining a Transformer from scratch with KerasHub
- Simple MNIST convnet
- Image classification from scratch using Keras 3
- Image Classification with KerasHub

```
import numpy as np
import matplotlib.pyplot as plt

fuel_efficiency = np.array([22, 28, 32, 36])

car_models = ["Model A", "Model B", "Model C", "Model D"]

average_efficiency = np.mean(fuel_efficiency)

percentage_improvement = ((fuel_efficiency[3] - fuel_efficiency[0]) / fuel_efficiency[0]) * 100

print("Average Fuel Efficiency:", average_efficiency)
print("Percentage Improvement from Model A to Model D: {:.2f}%".format(percentage_improvement))

plt.figure(figsize=(8,5))

bars = plt.bar(car_models, fuel_efficiency)

bars[3].set_color("orange")

for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height + 0.5,
        f"{height:.1f}",
        ha="center"
    )

plt.title("Fuel Efficiency of Car Models")
plt.xlabel("Car Models")
plt.ylabel("Fuel Efficiency (MPG)")
plt.grid(axis="y", linestyle="--", alpha=0.7)

plt.show()
```

```
Average Fuel Efficiency: 29.5
Percentage Improvement from Model A to Model D: 63.64%
```